I2C Communication Function

I2C Communication Function
- 1. GPIO Pin Diagram
- 2、I2C Testing
  - 2.1、Installing Dependencies
  - 2.2、I2C Devices
    - 2.2.1 Querying the I2C Bus
    - 2.2.2 Querying I2C Devices
  - 2.3 Running the Program
- 3. Experimental Results

1. GPIO Pin Diagram

Use a 0.91-inch OLED screen (not included) to test the I2C communication function. Connect the pins as shown below:

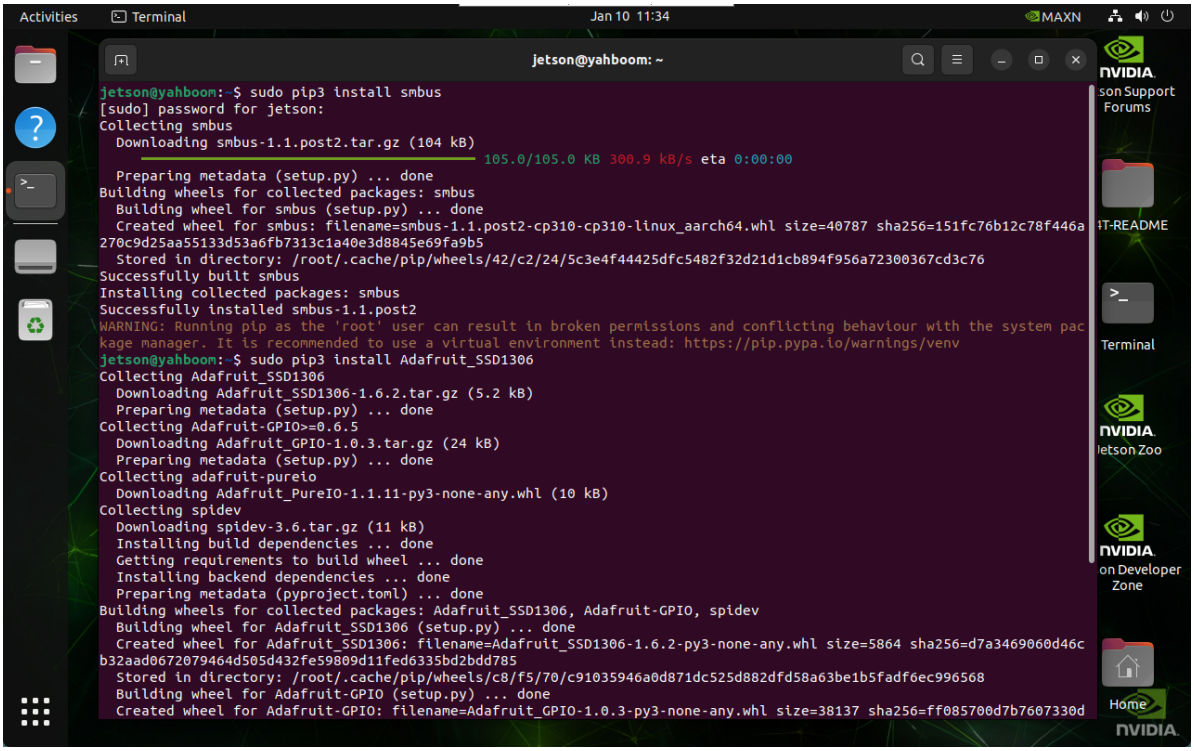
Caution: Do not connect incorrectly or cause a short circuit. Errors may damage the motherboard hardware!

GPIO.BOARD	OLED screen pins
1	VCC
3	SDA
5	SCL
9	GND

2、I2C Testing

2.1、Installing Dependencies

```
sudo pip3 install smbus
sudo pip3 install Adafruit_SSD1306
```



2.2、I2C Devices

During normal development, we need to find the device bus and device address of the I2C device.

2.2.1 Querying the I2C Bus

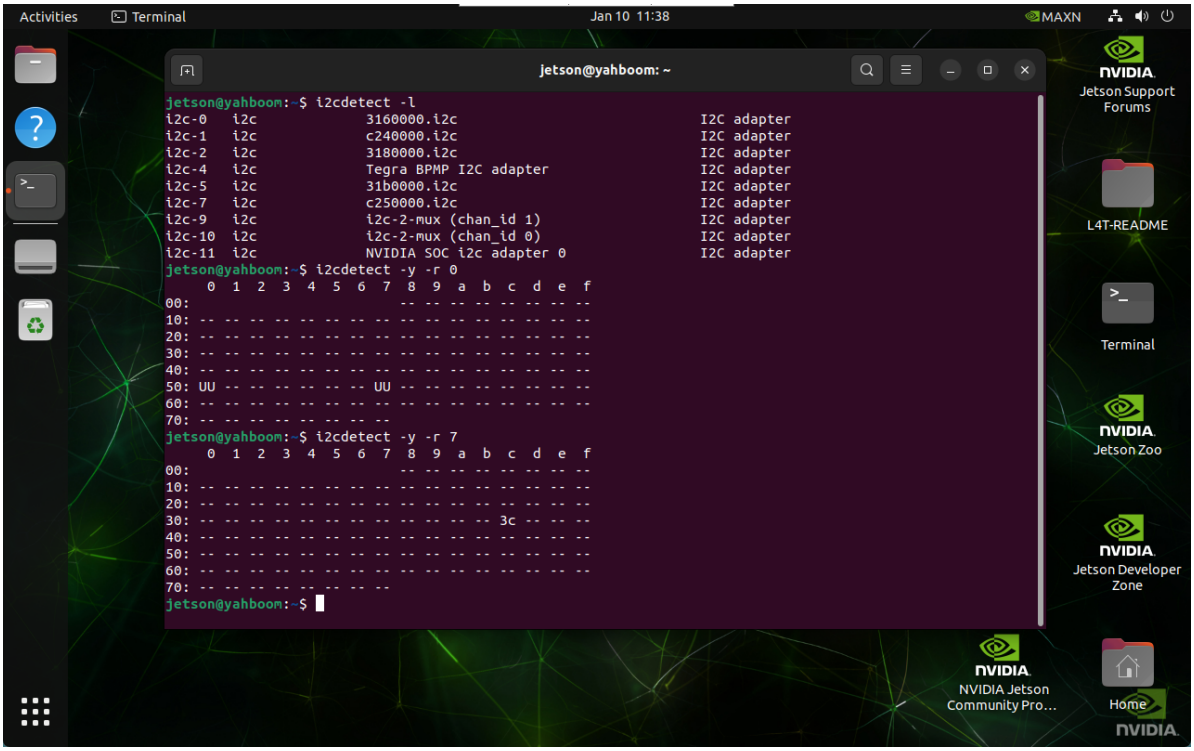
The following command in the terminal will list all buses for the device:

```
i2cdetect -l
```

2.2.2 Querying I2C Devices

The following command in the terminal will list the I2C devices under a specified bus: The I2C address corresponding to OLED is 0x3c

```
i2cdetect -y -r *
```



2.3 Running the Program

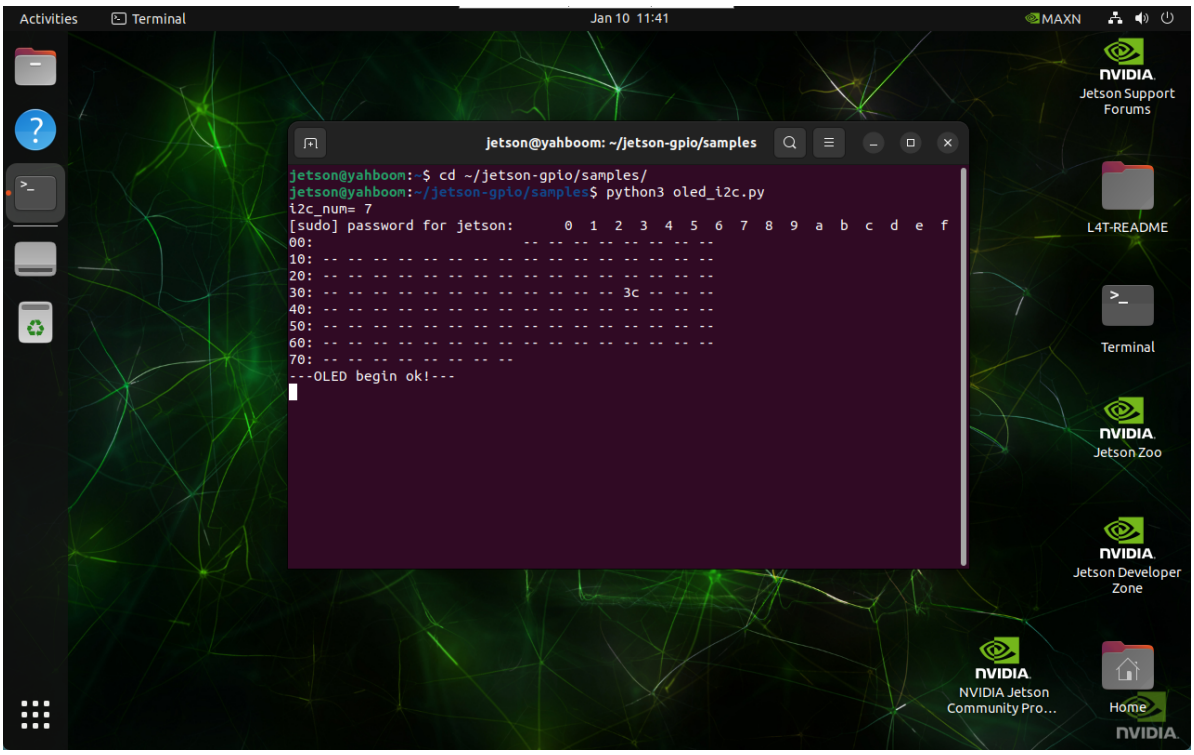
`oled_i2c.py` is not included in the `jetson-gpio` library:

```
cd ~/jetson-gpio/samples/
```

```
python3 oled_i2c.py
```

3. Experimental Results

After starting the program, the OLED will display system information such as CPU usage, system time, and memory usage:



NVIDIA
CPU:3% 18:21:51
RAM:3%->51.4GB
SDC:53%->450.0GB
IP:A:192.168.10.99
SSD1306 I2C V116